

Redteam Infrastructure using Terraform and Ansible

Overview

This lab is used for automatic creation of redteam infrastructure, including C2 server, HTTP redirector and testing environments like AD server and terminals.

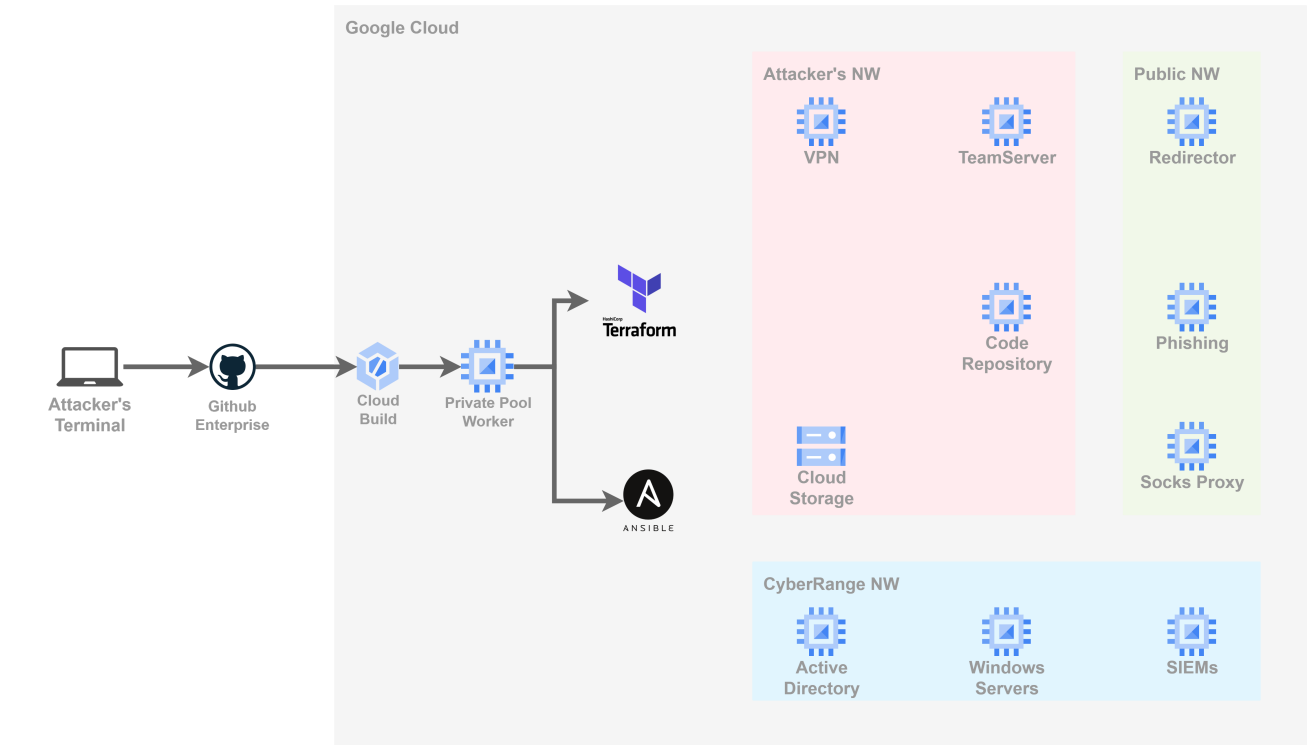
Warning: This lab is still early PoC stage and there are lots of hardcoded credentials. Please DO NOT use this lab in a production environment.

Functions

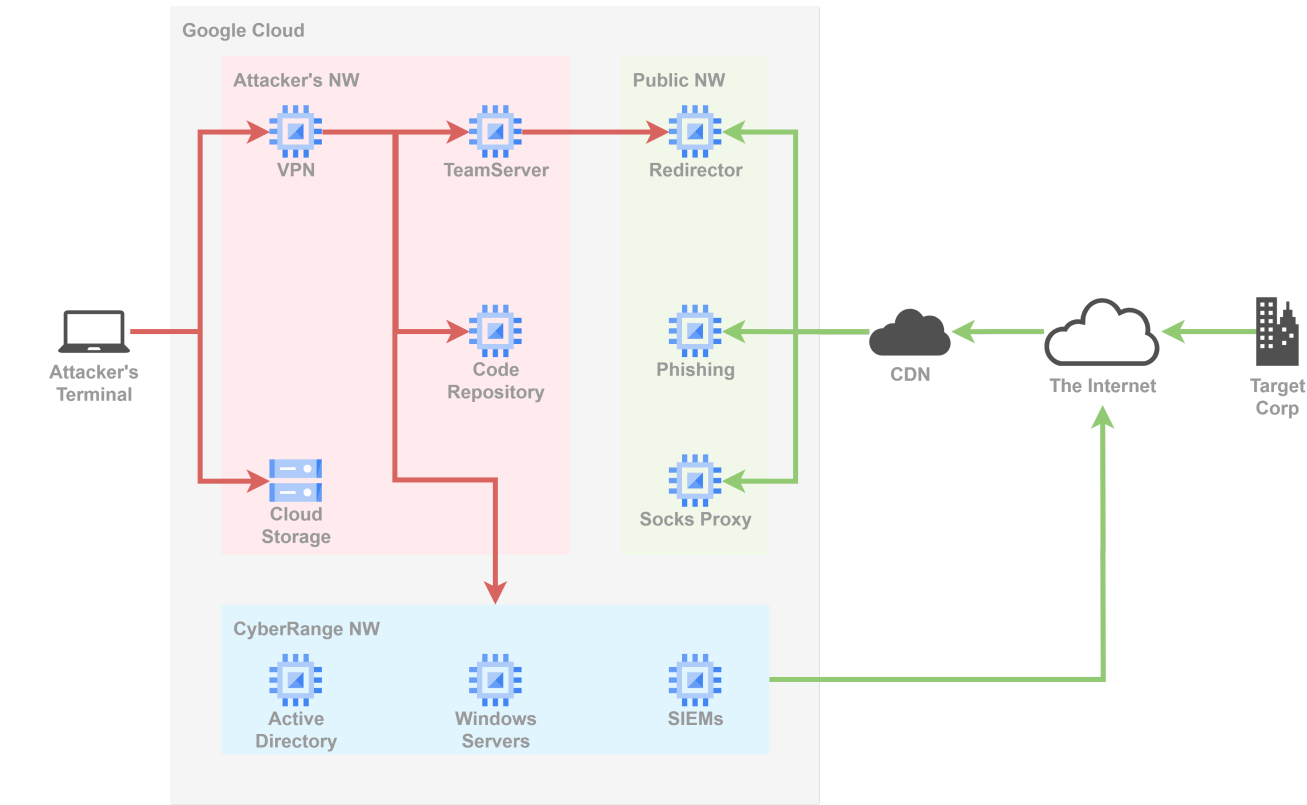
Name	Functions	Components
Teamserver	MultiPlayer	CobaltStrike
	Payload Generation	Havoc
	Listeners (HTTP/HTTPS,SMB,DNS)	Sliver
	Customizable C2 Profiles	
Attacker's terminal	Connect teamserver	Windows11
	R&D	Kali Linux
		VMWare/VirtualBox
Recdirector	HTTP/S traffic redirection	Apache Web Server
	Controlled by RewriteRule in Apache	
	SSH reverse port forwarding	
	Restrict direct inbound access to the C2	
CDN	Proxy traffic to the redirectors	CloudFront
		Azure CDN
		CloudFrare
Socks proxy	Passing through firewalls using HTTP/HTTPS	Chisel
	Increase network communication speed	
VPN	Connection from attacker's terminal to C2	OpenVPN
	MFA	YubiKey/Google Authenticator
Phishing	Upload C2 agents	Apache Web Server
Code repository	Git	GitLab
Storage	Cloud-based object storage	GoogleCloud Storage
	Policy based protection	
CyberRange Testing	Testing operation R&D	Windows Server
		Active Directory
		Windows11
		AV/EDR
		SIEM

Diagrams

- Building Flow



- Networking



GitOps style implementation

The lab uses Google Cloud to host virtual machines and virtual networks.

In order to make the environment disposable, almost all the components are implemented using IaC(terraform and ansible) for automatic construction.

The codes is intended for continuous integration and delivery (CI/CD) pipeline on CloudBuild and Github.

Requirements (manual creation)

- Github repository which hosts these codes
- Google Cloud project
- Service Account which has all the privileges below
 - under construction
- Storage blob to store tfstate file
- VPC (name: redteam-vpc)
- VPC peering settings for cloudbuild private pool
- CloudBuild settings
 - Trigger: start building stage when pushed to the Github branch
 - private pool

Usage

To build the lab, push this repository to the Github branch.

Roadmap

- Eliminate hardcoded creds
- Eliminate manual creation parts described above
- Create CDNs
- Documentation